

Problem

A linear network has this transfer function

$$H(s) = \frac{7s^2 + s + 4}{s^3 + 8s^2 + 14s + 5}, \quad s = j\omega$$

Use *MATLAB* or equivalent to plot the magnitude and phase (in degrees) of the transfer function. Take $0.1 < \omega < 10$ rad/s.

Step-by-step solution

Step 1 of 3

The transfer function of the linear network is,

$$H(s) = \frac{7s^2 + s + 4}{s^3 + 8s^2 + 14s + 5}$$

Step 2 of 3

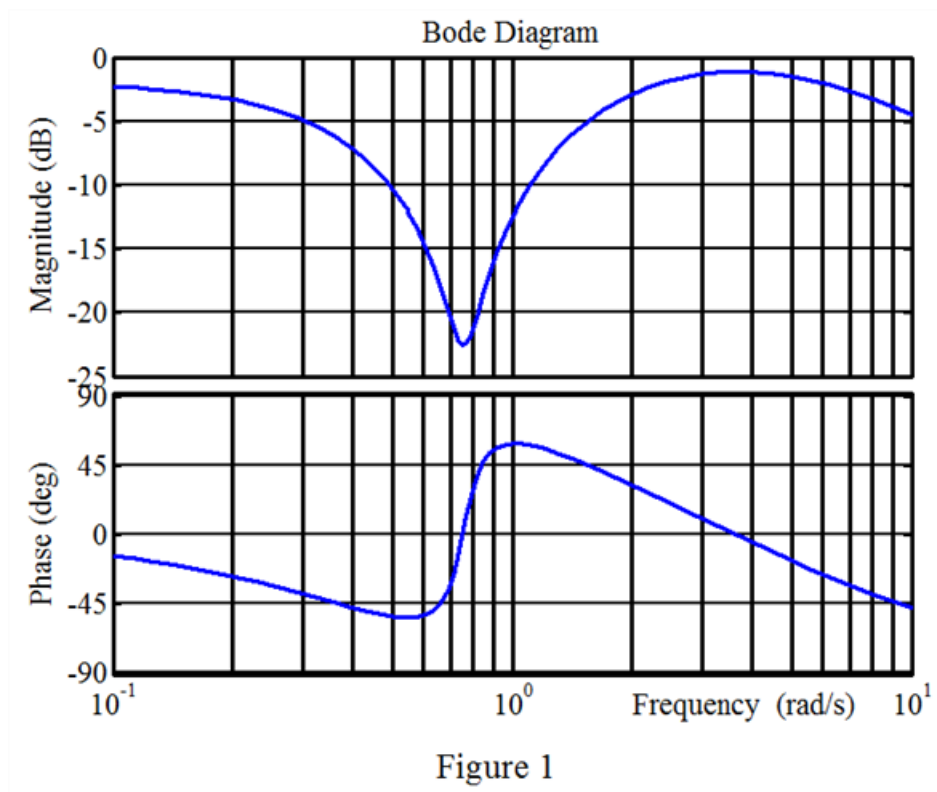
Sketch the magnitude and phase plots of linear network using MATLAB.

MATLAB code:

```
>> num=[7 1 4];  
>> den=[1 8 14 5];  
>> sys=tf(num,den);  
>> bode(sys,{0.1,10})
```

Step 3 of 3

Sketch the magnitude and phase plots of linear network.



Thus, the magnitude and phase plots of linear network are sketched as shown in Figure 1.